

GELEX: Generative AI-Hybrid System for Example-Based Learning

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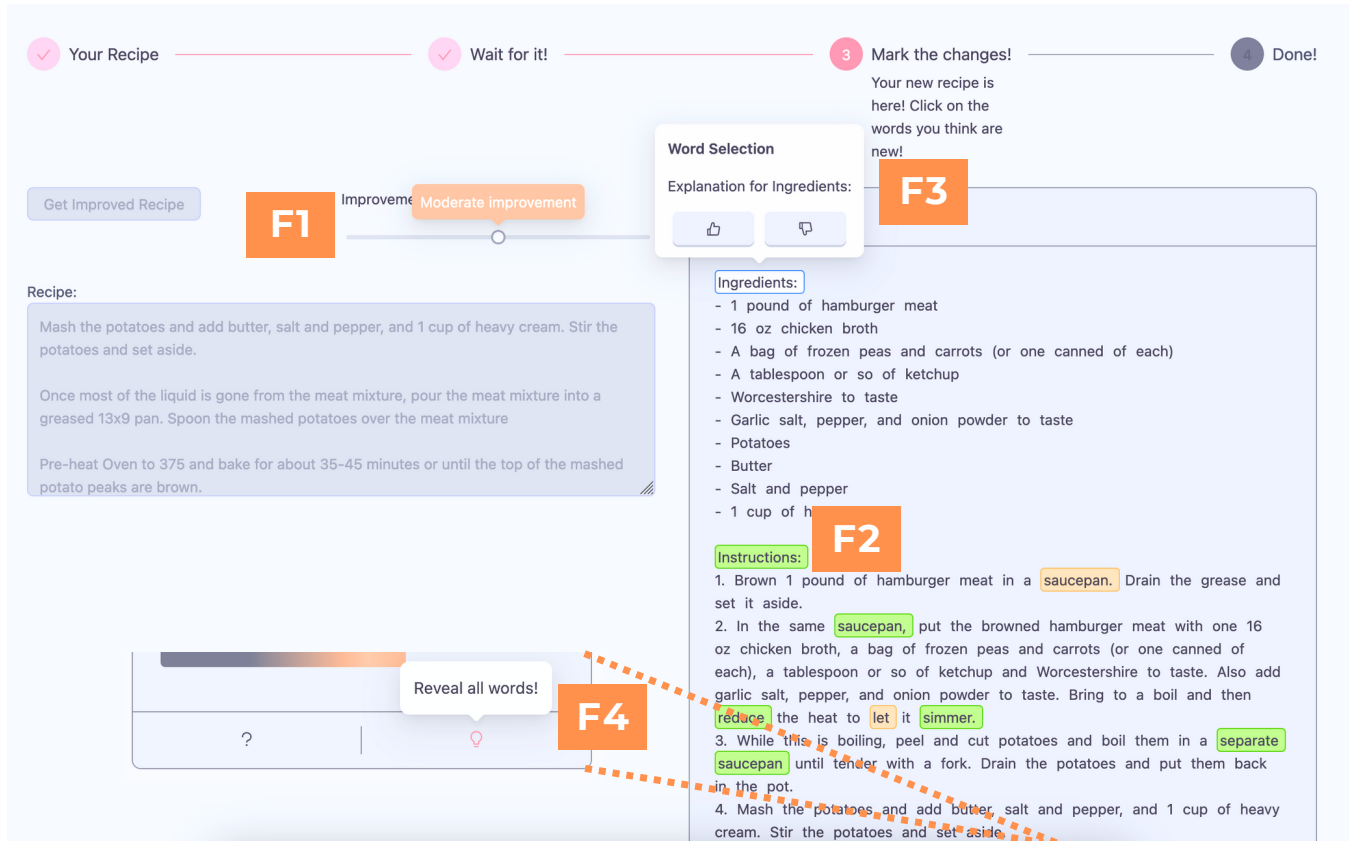


Figure 1: A learner requests feedback on a recipe specifying the improvement level (F1). Then, the learner receives a tailored example recipe and is prompted to actively process the example by highlighting the changes (F2) and critically assessing the modifications (F3) by seeing an explanation and deciding whether they want to keep or reject the changes. The learner can request hints or to reveal all changes (F4).

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ABSTRACT

Traditional example-based learning methods are often limited by static, expert-created content. Hence, they face challenges in scalability, engagement, and effectiveness, as some learners might struggle to relate to the examples or find them relevant. To address these challenges, we introduce GELEX (GEnerative-AI Learning through EXamples), a hybrid Artificial Intelligence (AI) system enhancing example-based learning by using large language models (LLMs). Our hybrid system incorporates mechanisms to control and evaluate the AI output, acknowledging and addressing the potential

factual inaccuracies of LLMs. We instantiate our system in the cooking domain. Our approach utilizes association rule mining on a large database of recipes to identify key patterns. When learners submit a recipe for feedback, a LLM enriches it by integrating these patterns. Then, learners are prompted to actively process the example by highlighting the changes and critically assessing the modifications. This strategy transforms traditional example-based learning into a dynamic, scalable, interactive educational tool.

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing; Machine learning**; • **Applied computing** → **Interactive learning environments**.

KEYWORDS

Example-based Learning, Procedural Writing, Generative AI

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1 INTRODUCTION

Learning from examples has been identified as an effective approach for learners to *borrow* knowledge from others [6, 35]. It enables learners to identify and extract general rules and apply them in similar future situations. This learning method has been widely studied in highly structured domains with single algorithmic solutions such as mathematics and physics [11, 35, 36].

In recent years, the applicability of example-based learning has expanded to include heuristic domains, such as writing tasks, which are characterized by the absence of precise, predefined solutions [26]. In these domains, examples serve a critical role as they provide a framework within which the rules, that are not as easily taught through direct instruction, can be inferred by the learners. Additionally, these examples are often paired with instructional explanations that highlight the essential attributes that learners should focus on. One of the limitations of this approach is the reliance on expert-created examples, limiting the availability of such learning resources, thus making example-based learning less accessible to a broader audience. Consequently, the created examples and instructions are mostly static, and hence not considering the varying levels of proficiency among learners, potentially hindering the effectiveness of the learning experience.

Indeed, adaptive feedback in writing tasks has been valued positively in various domains. For example, [5, 9, 10, 14, 15, 37, 38, 43] showcase the potential of machine learning (ML) and natural language processing (NLP) technologies in providing personalized feedback and recommendations for different writing tasks. These adaptive feedback writing tools encompass a range of genres, including argumentative writing (e.g., [21, 38, 40]), creative writing (e.g., [9]), scientific writing (e.g., [16]) and grammar feedback in language learning (e.g., [14, 15, 43]). However, while the aforementioned existing tools provide personalized suggestions, the majority

of them do not provide the learners with ideas or examples on how to implement these suggestions.

Recent advances in generative AI, in particular large language models (LLMs) can be leveraged to address this challenge by providing personalized content. However, the propensity of LLMs to produce factually incorrect or "hallucinated" content requires cautious application [44]. A hybrid model that combines the wisdom of the crowd with controlled modifications by LLMs therefore has the potential to mitigate the shortcomings mentioned previously while providing grounded examples.

In this paper, we present GELEX (GEnerative-AI Learning through EXamples), a hybrid AI system for example-based learning instantiated in the cooking domain. Our system provides adaptive feedback on learners' recipes by showing them a tailored example recipe (see Figure 1). It is designed for learning settings at scale, i.e. where learners have limited or no access to trained instructors, for example, at scale. Specifically we rely on a 'wisdom of the crowd' approach where, we apply association rule mining to an extensive database of over 2 million recipes to find out the underlying patterns majority of people follow. When a learner writes a recipe and requests feedback, a LLM (GPT-4) is used to enhance the learner's recipe by incorporating the identified patterns. By engaging learners in identifying and reflecting on these modifications, GELEX promotes active processing and fosters critical thinking. The system exemplifies the potential of integrating generative AI in educational content creation, marking a significant step forward in the development of interactive, personalized learning experiences.

2 RELATED WORK

In example-based learning, the examples are often selected (or created) by the instructors, who then enrich them with explanations or prompts. Furthermore, example-based learning tends to be static, i.e. all learners see the same examples and explanations regardless of their previous knowledge or performance [25]. Providing personalized examples fitting to the context of the learner is time-consuming and hence limits the scalability of this learning mode. In comparison to conventional learning systems in which all learners receive the same content, adaptive learning systems provide targeted content (e.g., feedback [2], hints [22]) depending on learners' interests and needs [23]. Given our use case of personalized example-based learning, our study was mainly inspired by work on providing adaptive feedback (i.e., in our case the instructional explanations for the example) and on providing adaptive content (i.e., in our case the example) in the context of heuristic domains.

2.1 Adaptive Feedback

In contrast to generic feedback, adaptive feedback is dynamic, which means that different learners will receive different information [4]. While there is a range of research on providing adaptive feedback and hints in structured domains such as mathematics [22], programming [2], or learning factual knowledge [30], less work has focused on giving automated fine-grained suggestions and examples on how to implement these suggestions in heuristic domains (e.g., compositional writing).

Existing NLP-based writing support tools often provide holistic feedback on higher-level properties of the text such as grammar

mistakes, fluency, or coherence (e.g., *Grammarly* [18]). When providing more fine-grained feedback, one approach is to visualize students' revision history. *ArgRewrite* [1] provides students with a comparison of two of their own drafts, annotated with detailed (at the sentence and sub-sentence level) and categorized revision suggestions. Shibani [33] automatically generate revision graphs to visualize multiple revision patterns.

Another approach to fine-grained adaptive feedback is the use of an underlying domain-specific structure. In *ArguLens* [41], the authors enrich discussions and feedback in open-source communities by imposing an argumentation-enhanced representation (decomposition into argumentation components and standpoint) on users' texts. Moreover, *AL* [38] provides users with an overview of the structure of their text and summary scores. Similarly, *ELEA* [39], a system for supporting empathy in peer-review scenarios, gives feedback in terms of summary statistics on the cognitive and emotional empathy structures of students' texts. While all the mentioned systems provide automated detailed feedback, they do not give suggestions or examples on how to fix the mistakes.

Giving implementation examples has proven effective in structured domains (e.g., programming [2]). Thus, different from existing adaptive feedback systems for heuristic domains, GELEX combines automated fine-grained and personalized feedback with an example, pointing out how to implement the feedback.

2.2 Adaptive Content

In content level adaptation, the learning objects (e.g., examples, tasks) are selected and adapted based on the knowledge of the user [23]. While learning objects in structured learning domains can be automatically created in a rule-based manner (e.g., arithmetic [13], algebra [19], creating and selecting learning objects matching the expertise of the learner is a challenge in heuristic domains. Previous work has relied on experts or existing textbooks to create examples (e.g., computer animations [12], science passages [42], geometry problems [31]). However this approach to content generation does not scale, especially when personalization is envisioned.

One approach to automatically generate adaptive content is the completion of student text. The *CoAuthor* system [17] uses transformer-based language models (LLMs) to enhance argumentative and creative writing by suggesting example sentences for the continuation of their user's text. More recently, [8] analyzed the capabilities of GPT models to produce written texts such as narrative fragments to enable adaptability in students' learning pathways.

With GELEX, we aim to scale *personalized* example-based learning with automation. In contrast to existing work, we do not provide a perfect expert-created example. Instead, we allow the user to choose the desired level of improvement. Based on prior research on the discouraging effect of perceived peer excellence [28], we hypothesize that showing attainable examples will be better for learning.

3 GELEX - HYBRID AI SYSTEM

Our system GELEX is designed to provide personalized examples at scale and at the same time promote user engagement and active

learning. In the following, we describe the design and functionalities of GELEX as well as the pipeline for generating personalized examples. The application is accessible at: <https://go.epfl.ch/gelex>.

3.1 Design

The user interface of GELEX is illustrated in Figure 1. Initially, users are guided through the different functionalities to understand the application's components. After this preliminary stage, users proceed to enter their recipe into a text editor. The system provides an adjustable slider, which allows the user to set the desired level of improvement (F1), thereby customizing the system's number of changes to suit individual preferences and expertise. Next, when the user requests feedback by clicking on *Get Improved Recipe*, they are presented with an example recipe on the right side panel. The generated recipe is similar to the users' recipe, but includes modifications to improve the recipe. The system then prompts users to identify modifications at a granular level (F2), either sentence-wise or word-wise, with the ability to toggle between these options as needed. Users must click on the changes (words or sentences) that have been modified. When selecting a new element, an explanation for the change appears and the users are prompted to accept (the element will be highlighted in green) or reject (the element will be highlighted in red) the changes (F3). Besides promoting user engagement, the highlighting and feedback features serve a dual purpose. Firstly, it fosters an active learning environment where users are encouraged to critically evaluate and understand each proposed modification in relation to their existing knowledge and recipe intention. This practice aligns with educational theories that advocate for active participation in the learning process [29]. Secondly, the user's decisions to accept or reject modifications provide feedback to the system. This information refines the algorithm's pattern recognition capabilities, enhancing the system's performance. Consequently, the database of suggestions with initial patterns adapts and improves through ongoing user interaction.

Lastly, to address the cognitive load and maintain a learning-efficient environment [7, 34], the system offers a feature that allows users to uncover modifications on demand (F4). Users may choose to reveal changes one at a time or all at once, according to their individual needs. This element of the system is designed to support users of varying levels of expertise, ensuring that the learning process remains user-friendly and adaptable to individual needs.

3.2 Example Generation Pipeline

To provide the learners with tailored examples, we propose the multi-step pipeline depicted in Figure 2. The pipeline consists of an offline and an online part, which are each divided into several steps. In the offline part, we first select the best-performing prompt type based on the quantitative and qualitative evaluation of the outputs of multiple prompts (detailed in Section 3.2.1). Next, we extract the data-driven rules from a large database. As the name implies, the online part happens in real-time while using the tool. In this part, the learner's recipe is passed to the *Control Center* to identify the missing patterns and generate a specific prompt. Then, the LLM is prompted to obtain the improved recipe which is evaluated in the *Evaluation Center*. A notable feature of our pipeline is its modular design, where each segment operates independently. This

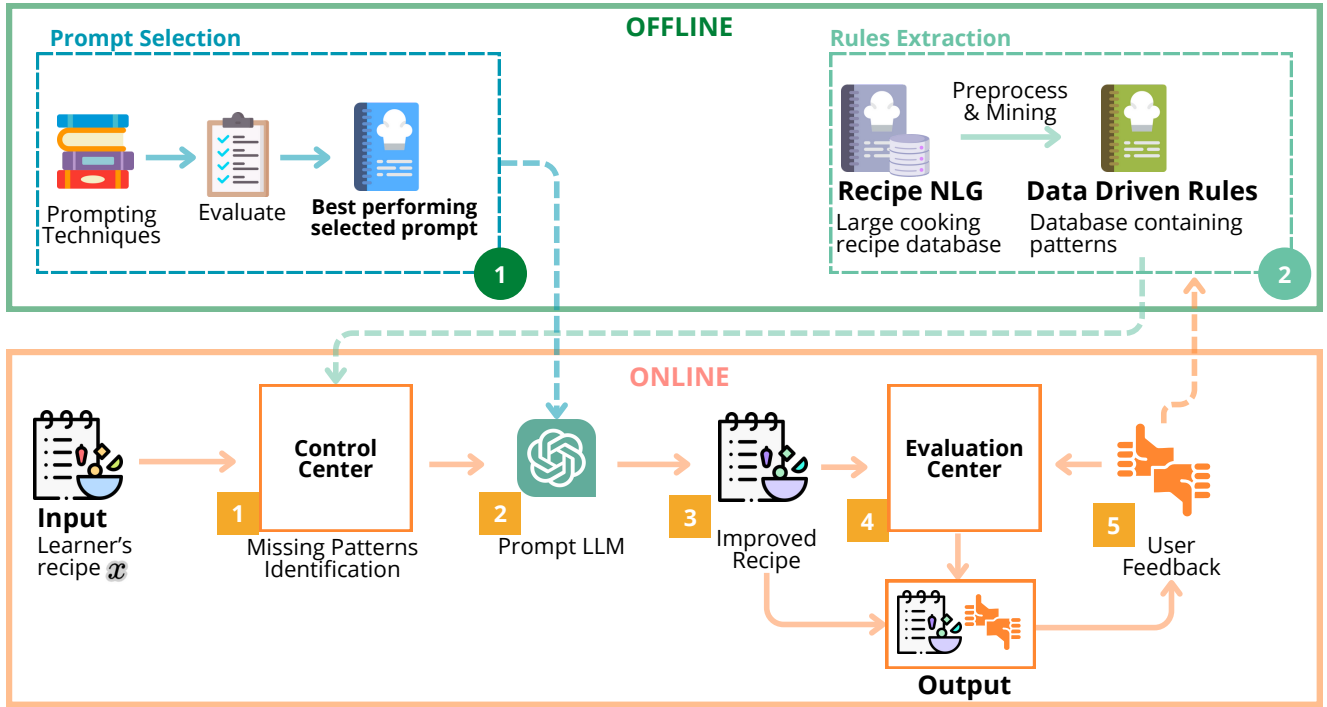


Figure 2: Diagram of our hybrid AI system consisting of two phases. In the offline phase, ① the best-performing prompt type is selected and ② the data-driven rules are extracted. The online phase takes as input the user’s recipe and ① identifies the missing patterns in the *Control Center*; ② using the missing patterns/rules, a specific prompt is generated according to the selected prompt type and ③ the improved recipe is retrieved from an LLM. ④ The *Evaluation Center* then identifies the changes and enables the highlighting for the user. Lastly, ⑤ the user gives feedback on the changes, which is returned to the *Evaluation Center* to adapt the relevance of the patterns for future iterations.

autonomy allows for the independent optimization of each step, facilitating focused and efficient enhancements within the pipeline. The source code is available at <https://github.com/epfl-ml4ed/gelex>.

3.2.1 Offline Pipeline. The *offline pipeline* consists of two main parts. In the **Prompt Selection** part, a prompt is constructed to instruct the chosen LLM to generate a recipe that follows a given set of rules. In the **Rules Extraction** part an initial set of rules is generated, based on the common patterns people follow when cooking.

Prompt Selection. In order to prompt a LLM to generate a recipe that follows a given set of rules, we explored various techniques. Our exploration of prompting techniques, influenced by the strategies outlined at PromptingGuide.ai, aimed to refine the process of recipe generation through generative AI. We used GPT-4 to try several methods and determined the most effective approach to enhance recipes according to predefined rules. The methods we evaluated were as follows:

- **Zero-shot Prompting:** This method involves presenting the LLM with a single recipe and a set of rules, and instructing the model to enhance the recipe by applying these rules.

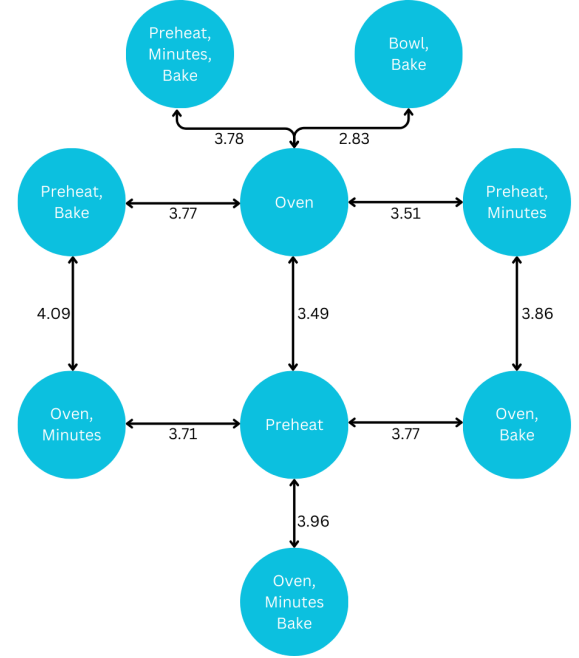
- **Few-Shot Prompting:** In contrast, Few-shot Prompting relies on the concept of providing the LLM with several correct examples. The model is then guided to discern modifications and replicate the examples. In our specific case, we provide the LLM with several examples of pairs of recipes. One recipe playing the role of the ‘original’ recipe, submitted by the user and to be improved by the LLM, the other playing the role of the ‘improved’ recipe, i.e. the recipe returned to the user/the output of the LLM. To be able to employ this technique, we needed pairs of “original” and “improved” recipes. To generate these pairs, we considered the recipes taken from RecipeNLG to already be improved, then we created the “non-improved” (i.e. original) recipes by omitting certain identified patterns.
- **Constraint-Based Prompting:** This technique involves defining explicit constraints that the LLM must adhere to during the recipe modification process. We applied this method in both the zero-shot and few-shot scenarios to observe the impact of strict guidelines on the generation outcome.

We evaluated the prompts¹ in two stages. The first stage was a human evaluation with two annotators who rated the output of the prompts according to the perceived relevance. Relevance was defined as the changes that the human annotators would keep and judge as appropriate and pertinent. In a second stage, the best-perceived prompts were systematically evaluated using recipes generated by real users.

Rules Extraction. To derive initial patterns, which will continuously change with user feedback, we followed a data-driven approach leveraging a dataset of over two million recipes, RecipeNLG [3]. We employed association rule mining (ARM) to identify recurrent patterns, correlations, and associations in the dataset. Specifically, we executed ARM on the Instructions column, which we preprocessed by stemming, stripping, and stopword removal to ensure data uniformity and relevance. The extraction of rules was conducted using the Instructions columns of the RecipeNLG database for two main reasons. Firstly, these patterns are more straightforward to align with the user’s recipe when considering it in its entirety. This holistic approach allows for more effective matching of patterns. Secondly, focusing on the Instructions column circumvents the potential challenge of parsing only the ingredients list from the user’s recipe submission. In many cases, users may not distinctly separate the ingredients from the rest of the recipe text, making it challenging to isolate and match patterns solely from the ingredients list. Consequently, relying on the Instructions column ensures a more consistent and reliable rule extraction process.

We considered each recipe as a *transaction* and calculated the confidence (reliability of a rule, measuring how often items in Y appear in transactions containing X); lift (strength of a rule compared to the expected likelihood of the antecedent and consequent appearing together by chance) and Zhang’s Metric (normalized assessment of the certainty and directionality of a rule’s implication) through ARM. This process yield a comprehensive set of rules, each accompanied by the aforementioned metrics. These statistics show the likelihood of the occurrence of item C given the presence of item A within a recipe. Figure 3a illustrates the interaction graph of “oven”. The nodes are the set of words and the edges contain the lift scores. The lift score measures how much more often the antecedent and consequent of a rule $A \rightarrow C$ occur together than we would expect if they were statistically independent. We can for example conclude that having the term “oven” in the recipe makes it 3.77 times more likely that also the terms “preheat” and “bake”.

Utilizing these metrics, we pinpoint and propose enhancements to recipes submitted by users. For a given user’s recipe, we identify the top k rules $A \rightarrow C$, based on the chosen metric, ensuring that while the antecedent is present, the consequent is absent. For example, Figure 3b shows the rules for Jello recipes. As shown in the figure, for the word “dissolve”, the neighbors are “boiling” and “water”. This suggests that if a user mentions that the Jello should be dissolved, the user should also refer to “boiling water”. The retrieved rules form the foundation of our recommendations for recipe improvement.



(a) Example rules with lift scores on the edges.

```
Recipe: frozenset({'water'})
Neighbors: {'dissolve', 'add', 'chill', 'boiling', 'jello'}

Recipe: frozenset({'dissolve'})
Neighbors: {'add', 'water', 'pineapple', 'boiling'}

Recipe: frozenset({'add', 'water'})
Neighbors: {'dissolve', 'chill', 'pineapple', 'boiling', 'jello'}

Recipe: frozenset({'boiling'})
Neighbors: {'dissolve', 'add', 'chill', 'water', 'pour', 'pineapple'}

Recipe: frozenset({'cream'})
Neighbors: {'add', 'sour', 'water', 'cheese'}
```

(b) ARM run on Jello recipes.

Figure 3: Visualizations of ARM rules in graph (a) and written form (b).

3.2.2 Online Pipeline. In the online pipeline, the learner’s recipe is processed and an improved example recipe is generated in real-time. As shown in the interface in Figure 1, a user enters their recipe and selects the desired level of improvement. The improvement levels are minimal, slight, moderate, significant, and major, corresponding to rule counts 1,3,5,10, and 30, respectively. Rule count 30 results in a warning pop-up informing the user of potential mistakes. The *online pipeline* takes the recipe and parameters as input and produces the output through five steps.

In a first step, the recipe is processed in the *Control Center* to match against the patterns identified from the RecipeNLG database. The most likely patterns that are missing are then pinpointed in a prompt. Using the selected prompting method, an LLM prompt is created to generate the example recipe. The recipes is formatted in order that the user can highlight the changes and explanations are added. Next, the user is presented with the enhanced recipe and is

¹All prompts are available at <https://github.com/epfl-ml4ed/gelex/blob/master/docs/exploratory-prompts.txt>

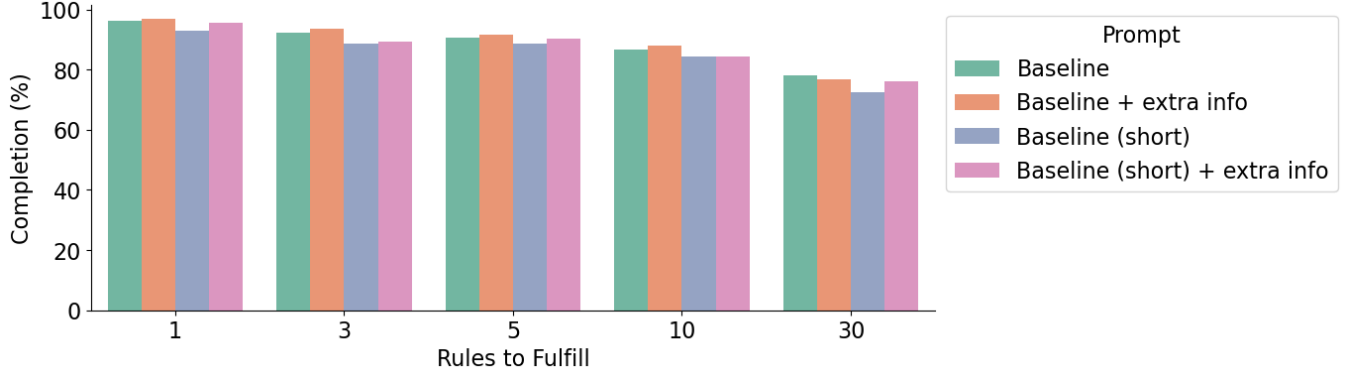


Figure 4: Comparison of percentages of fulfilled rules per prompt and number of rules given.

asked to identify the changes at a detailed level, choosing to view them by individual words or sentences. Finally, upon identifying a change, the user receives an explanation, providing insight into the rationale behind each modification. The user then decides whether to accept each change. This decision in turn informs the pattern database: accepting a change signals a successful application of a pattern, raising its importance in the database, while rejecting it could lower the pattern’s relevance. This process improves the recipe and also refines the pattern-recognition system, creating a cycle of continuous enhancement.

The generation of the improved recipes (steps 1-3 of the online pipeline, see Figure 2) given a metric (confidence, lift, or Zhang’s metric) follows a structured algorithm. The user’s recipe undergoes preprocessing to remove stop words, punctuation and is then stemmed. This processed recipe is denoted as R . The improved recipe R' is then initialized as $R' = R$. The rules from our dataset X , pre-sorted according to the chosen metric, are then iterated. The algorithm is delineated as follows:

Algorithm 1 Generation of Recipe Suggestions

- 1: **while** number of new suggestions < desired improvement level **do**
 - 2: Extract the next rule $A \rightarrow C$ from X
 - 3: **if** $A \in R$ and $C \notin R'$ **then**
 - 4: Add C as a suggestion; update R' as $R' = R' \cup C$
 - 5: **end if**
 - 6: **end while**
 - 7: **Note:**
 - (1) C is compared with R' because if a different $A' \in X$ suggests the same C , it maintains a one-to-one mapping for each $A \rightarrow C$, facilitating a clearer explanation for the addition of C and ensuring a linear increase in suggestions with the level of desired improvement.
 - (2) A is compared with R as the user might reject suggested improvements (i.e., C), and thus, these are not used to derive new suggestions.
-

This algorithm ensures that the process of generating improved recipes is systematic, transparent, and aligned with the user’s desired level of recipe enhancement. Our methodology facilitates the

generation of suggestions that are easily interpretable by users (e.g., since you had the word ‘oven’ we suggest ‘preheat’). Moreover, users’ feedback can be incorporated as a penalty or bonus when sorting the available rules to be chosen.

4 INITIAL EXPERIMENTAL EVALUATION

To investigate the impact of the metric selection and prompting technique, we gathered 200 recipes from online participants as possible inputs from potential users.

4.1 Experimental Protocol

We recruited 200 paid participants on the Prolific platform with a mean age of 26.5 ($\sigma = 3$). 49% of the participants identified as female, 47% as male and 4% as others. We asked the participants to write a cooking recipe using the following task: *“It’s a Sunday afternoon and your best friend calls you with a very hectic voice: they need to prepare a dish for their family who is going to visit in the evening. Your friend asks you to provide them with a cooking recipe. Be aware that your friend has very little cooking experience and therefore you have to write the recipe as structured and understandable as possible.”*

4.2 Prompt Selection

In the process of refining our prompting techniques, we conducted human-in-the-loop evaluations. Two testers, unaware of the prompting technique applied, compared original recipes from the user study with their enhanced versions and ranked the output versions. This methodology helped us rank the perceived relevance of the prompting strategies. As a result of these evaluations, we found that zero-shot prompting with constraints was the most preferred. Moreover, our findings indicated that instructing the model not to generate “flavour text” is less effective than guiding it toward the desired content creation. In addition, reducing verbosity and focusing on the essential elements of the instruction facilitates a more direct and efficient communication of the task to the model. These tests also highlight the challenges associated with few-shot prompting in the context of complex topics like recipes, where the need for generalization from limited examples is less practical. The insights and generalizations derived from one recipe, say a baking recipe, might not directly apply to a burger recipe.

In a next step, we created variations of the zero-shot prompt. The best-performing prompt can be found in Appendix 5². We then evaluated them in the pipeline, varying the number of new rule suggestions (k) at set intervals: 1, 3, 5, 10, and 30. We used the 200 recipes provided by the users in the study. The criterion for effectiveness was the extent to which the suggested rules were actually incorporated into the output recipe.

Figure 4 shows the results of the evaluation. We observe an inverse relationship between the number of rules provided and the completion rate. As the volume of rules increased, the precision of the prompt correspondingly diminished, due to the prompt getting considerably longer, which led to a notable decrease in rule fulfillment. Specifically, a 20% decline in adherence was recorded when the model was prompted with a higher count of suggestions, underscoring the importance of prompt conciseness and specificity. Moreover, the baseline with extra information seems to consistently fulfill more rules than the other prompting strategies, thus that was the chosen strategy.

4.3 Metric Selection

In order to study how different metrics influence the chosen rules, we investigated which would be the top rules presented to each user given their recipe sorted by each algorithm. Similar to the previous experiment, the user-submitted recipes served as the input for this study. For each recipe, we identified the top ten rules generated by each metric using the algorithm described in subsection 3.2.2. The most prevalent suggestions from each metric are visualized in Figures 5, 6, and 7. In the following, we detail the findings and the implications they have on the rule selection process.

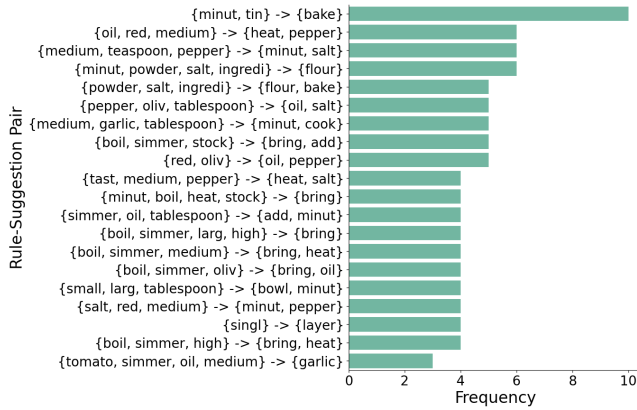


Figure 5: Confidence Metric Results.

Confidence. The selection of top suggestions through the confidence metric presupposes a preference for rules where the antecedent part is long, whereas the consequent part is short. As we get a larger antecedent, the subset of the recipes we consider gets smaller, hence the increase in probability. From Figure 5, it is evident that the antecedent part for each suggestion encompasses two

or three words, and not many recipes share common suggestions, i.e., each recipe gets almost a unique suggestion.

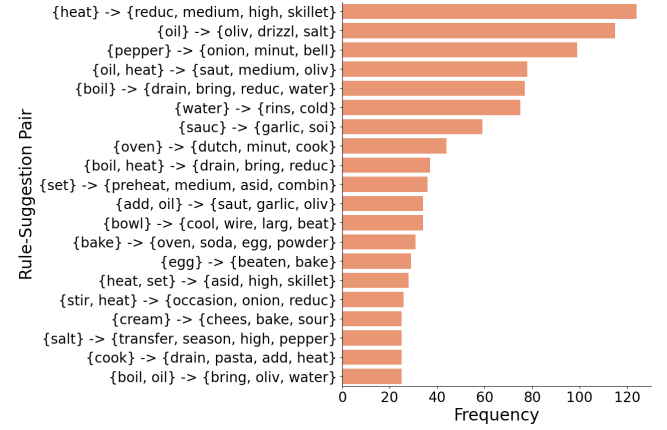


Figure 6: Zhang's Metric Results.

Zhang's Metric. Analyzing the antecedent-consequent pairs extracted using Zhang's Metric, depicted in Figure 6, we notice that the metric tends to select pairs where the antecedent contains a mix of specific and general terms, and the consequent is usually a specific action or ingredient that is significantly associated with the antecedent. For instance, the presence of "heat" as an antecedent is frequently associated with a variety of actions and tools, such as "reduc", "medium", "high", "skillet", which suggests that Zhang's Metric is sensitive to nuanced relationships where the occurrence of a common term like "heat" in the context of cooking is indicative of specific cooking actions and conditions. This sensitivity makes Zhang's Metric particularly useful for identifying rules that are not only frequent but also meaningful within the given context.

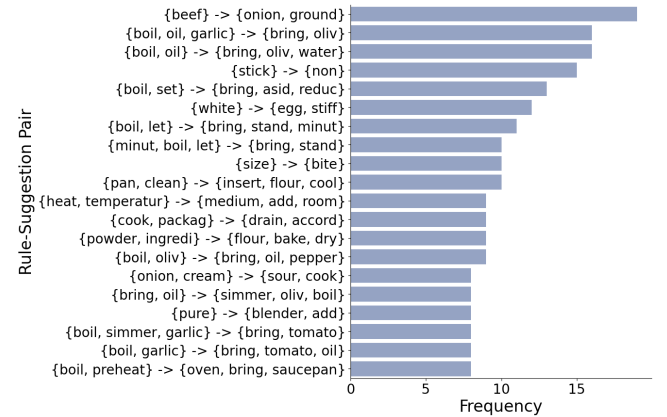


Figure 7: Lift Metric Results.

Lift. From Figure 7, we observe that the antecedent sets often contain ingredients or actions that are less common in the dataset, while the consequents are more commonly occurring items or actions. This suggests that when these less common antecedents are

²All the evaluated prompts are available at <https://github.com/epfl-mil4ed/gelex/blob/master/docs/prompts.txt>

present, there is a significant increase in the probability of the consequent occurring. The pairs extracted using Lift as the metric indicate strong and specific relationships; for example, the pairing of uncommon ingredients like “garlic” and “boil” with more general actions such as “bring” and “tomato” suggests that these specific conditions or ingredients are particularly predictive of the outcome, beyond what would be expected by chance alone. For this reason, we chose *lift* as the sorting metric.

5 DISCUSSION AND CONCLUSION

In this paper, we designed GELEX, an adaptive learning system for example-based learning based on generative AI. We conducted an initial experimental evaluation involving 200 participants to assess various prompting techniques and evaluate the different metrics (confidence, lift, and Zhang’s metric) in the rule selection process.

One of the main advantages of our design is that it can be employed to generate personalized content at scale. The current instantiation could for example be used by chef apprentices to improve their recipe writing skills [20]. In the future, the design and pipeline could be adapted to different domains (e.g., journal writing [27], designing worked examples for high school instruction [11], and argumentative writing [32]).

One important concern is the potential Western bias due to the RecipeNLG dataset’s composition. Future work should focus on diversifying the data sources. Incorporating recipes from various global cuisines would enhance the system’s versatility and cultural inclusivity. Additionally, the rules extraction model using ARM, while effective in many cases, sometimes oversimplifies complex relationships. Adopting more sophisticated models could address the current limitations in capturing the intricacies of language. For example, by gathering labels in the form of human preference and using Direct Preference Optimization (DPO) [24] to align the LLM’s generations with the users’ preferences. Moreover, in the future, the different components of GELEX and their effectiveness for learning should be assessed in controlled studies and in real classrooms.

In conclusion, GELEX leverages the capabilities of generative AI and, at the same time, addresses the potential inaccuracies inherent in LLMs by adopting a hybrid approach, enhancing the reliability of the content provided. Moreover, GELEX encourages learners to think critically by having to accept or reject the suggested changes. Our system represents a step forward in interactive and personalized learning experiences using hybrid AI systems.

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APPENDIX

In this section, we detail the best-performing prompt used in GELEX.

Prompt Used: Baseline + Extra Information

You are a recipe improvement assistant. The improvement will be done ONLY in the scope of rules. You will be given a recipe and a set of rules that it has already fulfilled. Note that this will just be a subset of all the rules that the recipe fulfills. The rules will be of the following shape: frozenset({{ 'word1', 'word2', ... }}) - This means that the words word1, word2, ... should be present somewhere in the recipe. Note that, these words aren't dependent on each other. Thus they don't have to appear in the same sentence, or in the same order that they are given. It just means they have to appear at least once somewhere in the recipe. The user will also give you a new set of rules that it has not fulfilled yet. You are responsible for rewriting the recipe. You have to make sure that the new recipe you write fulfills all the new rules while keeping all the details from the original recipe intact. Thus, you are to only add to the original recipe, and avoid removing anything from it. You are to only add something if it directly helps you fulfill the new rules. You'll write two parts, the first part is the Ingredients and Instructions. The second part is the explanation.

The first part will be wrapped between <RECIPE> and </RECIPE> tags. In this part include the ingredient portions in the list labelled Ingredients: and then the Instructions section as a numbered list.

The second part will be wrapped between <EXPLANATION> and </EXPLANATION> tags. In this part, explain why you made the changes you made.

So the output format is:

<RECIPE>

Ingredients:

- Ingredient 1
- Ingredient 2
- ...

Instructions:

- (1) Step 1
- (2) Step 2
- (3) ...

</RECIPE>

<EXPLANATION>

Your explanation here

</EXPLANATION>